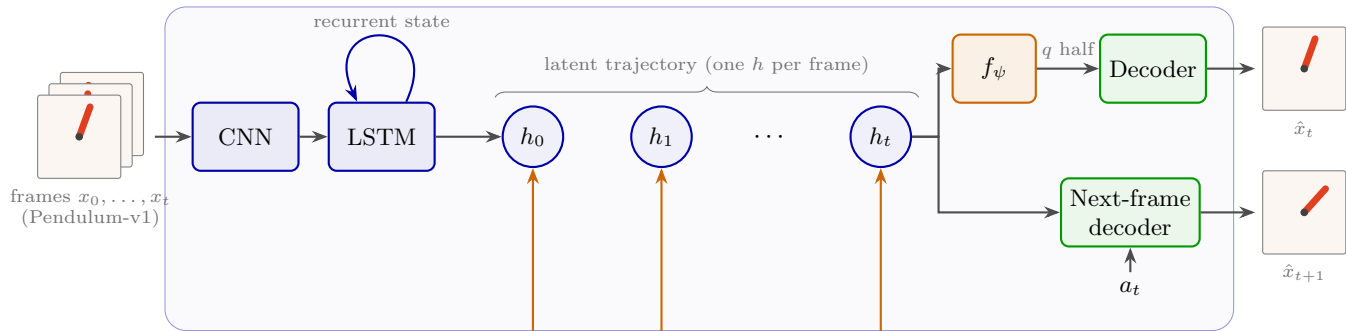
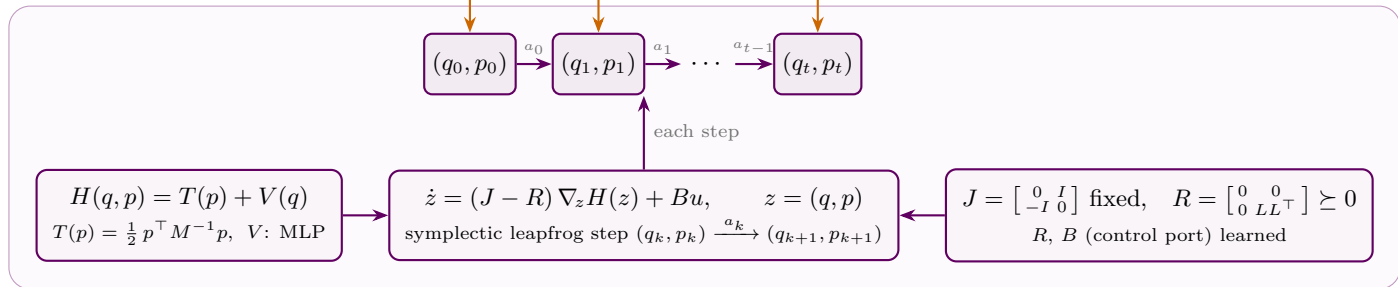


Phase 1 — LSTM autoencoder (pixel space)



Φ — normalizing flow (bijective)
 $(q_i, p_i) = \Phi(h_i), \quad h_i = \Phi^{-1}(q_i, p_i)$



Phase 2 — port-Hamiltonian dynamics (learned phase space)